

Emily Wall, Ph.D.

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Emory University, Computer Science Department
400 Dowman Drive, Suite W302K
Atlanta, GA 30322

www.emilywall.prof
emily.wall@emory.edu

SUMMARY

Prof. Emily Wall works in the areas of decision making, data visualization, and HCI. She develops computational strategies to characterize human limitations in decision making (e.g., cognitive bias) and designs interventions to promote socially responsible and reflective data analysis and decision making processes.

EDUCATION

- **Ph.D., Computer Science, Georgia Institute of Technology, USA (2020)**
Thesis: “Detecting and Mitigating Human Bias in Visual Analytics”
- **B.S., Computer Science and Mathematics, University of Georgia, USA (2015, *summa cum laude*)**

APPOINTMENTS

- Assistant Professor, Emory University, Atlanta, GA (June 2021-present)
- Postdoctoral Researcher, Northwestern University, Evanston, IL (June 2020-June 2021)

SELECTED GRANTS

- Promoting Metacognition in Visual Analytics (NSF CAREER Award)
 - Role: PI, Award Amount: \$643,965, Award Period: 2024-2029
- Advancing Visualization Decision Support through Mitigating Confirmation Bias (NSF IIS-HCC)
 - Role: PI, Award Amount: \$683,097 (Emory) of \$1,198,084 (total), Award Period: 2023-2027
- Toward Interventions for Equitable University Admissions with Visual Analytics (Emory Office of Provost: Racial Justice / Racial Equity)
 - Role: PI, Award Amount: \$148,850, Award Period: 2023-2026
- Developing Behavior Change Interventions for Responsible Data Science (University Research Committee)
 - Role: PI, Award Amount: \$30,000, Award Period: 2023-2024
- Developing an Immersive Cognitive Task for Expanding the Assessment of Memory in Pediatrics (Pediatric Research Alliance Pilot Grant)
 - Role: Co-I, Award Amount: \$50,000, Award Period: 2023-2024
- A Novel Web-based Decision Aid Tool for Reducing COVID-19 Vaccine Hesitancy (Healthcare Innovation Seed Grant)
 - Role: Co-I Award Amount: \$24,995, Award Period: 2022-2023

SELECTED AWARDS

- Fall 2023 Arts & Social Justice, fellow
 - Information Visualization (CS 441 / CS 541)
 - Faculty fellows at Emory are paired with a local Atlanta artist to redesign the curriculum of a course to incorporate arts and social justice. Worked with poet W.J. Lofton in Fall 2023 on an exhibit titled “Undoing the Plot: A Data-Driven Exhibit on Mass Incarceration and Police Brutality” (https://emilywall.github.io/vis/f23_asj.html) and several students additionally wrote and developed data-driven poems.
 - Spun into event at Atlanta Science Festival Spring 2025 that combines data, poetry, and social justice.
- 2022 Best Paper, *runner up*
 - Computer Graphics and Applications for paper “VisHiker’s Guide to Evaluation: Competing Considerations in Study Design”
- 2020 VGTC Best Dissertation, honorable mention
 - Thesis: “Detecting and Mitigating Human Bias in Visual Analytics”

TEACHING

- Emory CS 441/541: Information Visualization (Spring 2026 – 38 students; Spring 2025 – 39 students; Fall 2023 – 36 students; Fall 2022 – 41 students; Fall 2021 - 26 students)
- Emory CS 485/584: Human-Computer Interaction (Spring 2024 – 29 students; Spring 2023 – 34 students; Spring 2022 - 40 students)
- Emory CS 171: Introduction to Computer Science II (Fall 2026 – 26 students; Fall 2024 – 23 students)

SELECTED BOOKS AND CHAPTERS

- **Emily Wall**. “Human Bias in Visual Data Analysis (Synthesis Lectures on Visualization).” Springer Nature, 2026. *Open Access Link*: <https://link.springer.com/book/9783032093066>.
- **Emily Wall**, Leslie M. Blaha, Celeste Lyn Paul, Kristin Cook, and Alex Endert. “Four perspectives on human bias in visual analytics.” In *Cognitive biases in visualizations*, pp. 29-42. Springer, Cham, 2018.

SELECTED CONFERENCE AND JOURNAL PUBLICATIONS

- Mengyu Chen, Yixin Bai, Zheyuan Lin, and **Emily Wall**. “Less is More Reflective: Intervention Intensity and Metacognitive Disruption in Visual Analytics,” *IEEE Transactions on Visualization and Computer Graphics*, 2026.
- Shiyao Li, Akul Rana, Cindy Xiong Bearfield, and **Emily Wall**. “Confirmation Bias Shapes Outlier Identification and Removal in Data Cleaning with Scatterplots,” *IEEE Transactions on Visualization and Computer Graphics*, 2026.
- Thomas Davidson, Zihan Dai, Yalong Yang, **Emily Wall**, and Keke Wu. “Charting the Immersive Self: A Scoping Review of Immersive Personal Visualisation,” *IEEE Transactions on Visualization and Computer Graphics*, 2026.
- Aimen Gaba, **Emily Wall**, Tejas Ramkumar Babu, Yuriy Brun, Kyle WM. Hall, and Cindy Xiong Bearfield. “Bias, Accuracy, and Trust: Gender-Diverse Perspectives on Large Language Models,” *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FACCT)*, 2026.
- Jera White, Tej Shah, **Emily Wall**, Shivani A. Patel, Kevin Wu, Yuxuan Shi, Yuxi Zhang, Ziwei Dong, and Ambar Kulshreshtha. “Self-Administered Digital Vaccination Decision-Aids in Primary Care: A Pilot Study of COVID-19 Vaccination in Atlanta, GA,” *Digital Health*, 2026.
- Ziwei Dong, Keke Wu, Leilani Battle, and **Emily Wall**. “Evaluating Behavior Change Interventions for Responsible Data Science,” *CHI*, 2026.
- Shiyao Li, Roshini Deva, Arpit Narechania, Alireza Karduni, Cindy Xiong Bearfield, and **Emily Wall**. “Does a Picture Paint a Thousand Words? Using Visual and Textual Channels to Understand Attitudes and Beliefs,” *CHI*, 2026.
- Thomas Davidson, Ethan Lee, and **Emily Wall**. “Spotify Warped: Reshaping Personal Informatics via Music Listening, Casual Users, Passive Data, and Episodic Reflection,” *CHI*, 2026.
- Yanan Da, Yutong Bu, Yiling Li, and **Emily Wall**. “rutaBAGA: A Visualization Approach for Bias Awareness in University Admissions,” *Journal of Visualization and Interaction (JoVI)*, 2025.
- Yanan Da, Mengyu Chen, Ben Altschuler, Yutong Bu, and **Emily Wall**. “Visual Salience to Mitigate Gender Bias in Recommendation Letters,” In *IFIP Conference on Human-Computer Interaction*, 2025.
- Mengyu Chen, Andrew Yang, Aiden Min, Kristy Hamilton, and **Emily Wall**. “A Novel Lens on Metacognition in Visualization,” *CHI*, 2025.
- Shiyao Li, Thomas Davidson, Cindy Xiong-Bearfield, and **Emily Wall**. “Confirmation Bias: The Double-Edged Sword of Data Facts in Visual Data Communication,” *CHI*, 2025.
- Ziwei Dong, Teanna Barrett, Ameya Patil, Yuichi Shoda, Leilani Battle, and **Emily Wall**. “A Design Space of Behavior Change Interventions for Responsible Data Science,” *IUI*, 2025.
- Ziwei Dong, Ameya Patil, Yuichi Shoda, Leilani Battle, and **Emily Wall**. “Behavior Matters: An Alternative Perspective on Responsible Data Science,” *CSCW*, 2025.
- Mengyu Chen, Yijun Liu, and **Emily Wall**. “Unmasking Dunning-Kruger Effect in Visual Reasoning and Judgment,” *IEEE Transactions on Visualization and Computer Graphics*, 2024.
- Zhongzheng Xu and **Emily Wall**. “Exploring the Capability of LLMs in Performing Low-Level Visual Analytic Tasks on SVG Data Visualizations,” In *2024 IEEE Visualization Conference (VIS)*, 2024.
- [Best Paper] Jen Rogers, Marie Anastacio, Jürgen Bernard, Mehdi Chakhchoukh, Rebecca Faust, Andreas Kerren, Steffen Koch, Lars Kotthoff, Cagatay Turkay, and **Emily Wall**. “Visualization and Automation in Data Science: Exploring the Paradox of Humans-in-the-Loop,” *Workshop on Visualization in Data Science (VDS) at IEEE VIS*, 2024.
- Hongye An, Arpit Narechania, **Emily Wall**, and Kai Xu. “VITALITY 2: Reviewing Academic Literature Using Large Language Models,” *Workshop on Natural Language and Visualization (NLVIZ) at IEEE VIS*, 2024.
- **Emily Wall**, Laura Matzen, Mennatallah El-Assady, Peta Masters, Helia Hosseinpour, Alex Endert, Rita Borgo, Polo Chau, Adam Perer, Harald Schupp, Hendrik Strobel, and Lace Padilla. “Trust Junk and Evil Knobs: Calibrating Trust in AI Visualization,” *IEEE PacificVis Conference Proceedings*, 2024.

- Zhongzheng Xu, Kristin Williams, and **Emily Wall**. "Let's Get Vysical: Perceptual Accuracy in Visual and Tactile Encodings," *IEEE Transactions on Visualization and Computer Graphics*, pp. 16-20. 2023.
- Shrey Gupta, Alireza Karduni, and **Emily Wall**. "Belief Decay or Persistence? A Mixed-Method Study on Belief Movement Over Time," *Computer Graphics Forum (Proceedings of EuroVis)*, 2023.
- Thomas Davidson, **Emily Wall**, and Jonathan Mace. "A Qualitative Interview Study of Distributed Tracing Visualisation: A Characterisation of Challenges and Opportunities," *TVCG 2023*.
- Yanan Da, Mengyu (Bonnie) Chen, Ben Altschuler, Yutong (Audrey) Bu, and **Emily Wall**. "Visual Salience to Mitigate Gender Bias in Recommendation Letters," *Workshop on Visualization for Social Good at IEEE VIS*, 2023.
- [Best Paper, runner up] **Emily Wall**, Cindy Xiong, and Yea-Seul Kim. "VisHiker's Guide to Evaluation: Competing Considerations in Study Design," *IEEE Computer Graphics and Applications*, 2022.
- Shambhavi Mahajan, Mengyu (Bonnie) Chen, Alireza Karduni, Yea-Seul Kim, and **Emily Wall**. "VIBE: A Design Space for Visual Belief Elicitation in Data Journalism," *Computer Graphics Forum (Proceedings of EuroVis)*, 2022.
- Mengyu (Bonnie) Chen and **Emily Wall**. "Perception of Skill in Visual Problem Solving: An Analysis of Interactive Behaviors, Personality Traits, and the Dunning-Kruger Effect," *Workshop on Trust and Expertise in Visual Analytics at IEEE VIS*, 2022.
- **Emily Wall**, Arpit Narechania, Adam Coscia, Jamal Paden, and Alex Endert. "Left, right, and gender: Exploring interaction traces to mitigate human biases." *IEEE Transactions on Visualization and Computer Graphics*, 2021.
- Arpit Narechania, Alireza Karduni, Ryan Wesslen, and **Emily Wall**. "VITALITY: Promoting Serendipitous Discovery of Academic Literature with Transformers & Visual Analytics." *IEEE Transactions on Visualization and Computer Graphics*, 2021.
- Arpit Narechania, Adam Coscia, **Emily Wall**, and Alex Endert. "Lumos: Increasing awareness of analytic behavior during visual data analysis." *IEEE Transactions on Visualization and Computer Graphics*, 2021.
- Star Liu, **Emily Wall**, Shivani A. Patel, and Yubin Park. "COVID-19 Health Equity Dashboard-Addressing Vulnerable Populations." *Workshop on Visualization for Communication at IEEE VIS*, 2020.
- **Emily Wall**, Arup Arcalgud, Kuhu Gupta, and Andrew Jo. "A markov model of users' interactive behavior in scatterplots." In *2019 IEEE Visualization Conference (VIS)*, pp. 81-85. IEEE, 2019.
- **Emily Wall**, John Stasko, and Alex Endert. "Toward a design space for mitigating cognitive bias in vis." In *2019 IEEE Visualization Conference (VIS)*, pp. 111-115. IEEE, 2019.
- **Emily Wall**, Leslie Blaha, Celeste Paul, and Alex Endert. "A formative study of interactive bias metrics in visual analytics using anchoring bias." In *IFIP Conference on Human-Computer Interaction*, pp. 555-575. Springer, Cham, 2019.
- **Emily Wall**, Soroush Ghorashi, and Gonzalo Ramos. "Using expert patterns in assisted interactive machine learning: A study in machine teaching." In *IFIP Conference on Human-Computer Interaction*, pp. 578-599. Springer, Cham, 2019.
- **Emily Wall**, Meeshu Agnihotri, Laura Matzen, Kristin Divis, Michael Haass, Alex Endert, and John Stasko. "A heuristic approach to value-driven evaluation of visualizations." *IEEE transactions on visualization and computer graphics* vol. 25, no. 1 (2018): 491-500.
- **Emily Wall**, Leslie M. Blaha, Lyndsey Franklin, and Alex Endert. "Warning, bias may occur: A proposed approach to detecting cognitive bias in interactive visual analytics." In *2017 IEEE Conference on Visual Analytics Science and Technology (VAST)*, pp. 104-115. IEEE, 2017.
- **Emily Wall**, Subhajit Das, Ravish Chawla, Bharath Kalidindi, Eli T. Brown, and Alex Endert. "Podium: Ranking data using mixed-initiative visual analytics." *IEEE transactions on visualization and computer graphics* vol. 24, no. 1 (2017): 288-297.
- Sriram Karthik Badam, Zehua Zeng, **Emily Wall**, Alex Endert, and Niklas Elmqvist. "Supporting Team-First Visual Analytics through Group Activity Representations." In *Graphics Interface*, pp. 208-213. 2017.
- Bum Chul Kwon, Hannah Kim, **Emily Wall**, Jaegul Choo, Haesun Park, and Alex Endert. "Axisketcher: Interactive nonlinear axis mapping of visualizations through user drawings." *IEEE transactions on visualization and computer graphics* vol. 23, no. 1 (2016): 221-230.

MENTORING

- Postdoctoral Scholars:
 - Dr. Keke Wu (2024-2025) (after Emory: Assistant Professor at University of Maryland)
- PhD Advisees:
 - Sunggeol Oh (2026-present)
 - Thomas Davidson (2022-present)
 - Mengyu Chen (2021-present); Thesis: "Promoting Metacognitive Thinking within Visualization"

- Shiyao Li (2020-2026); Thesis: “The Role of Text, Annotations, and Visual Encodings in Shaping Confirmation Bias During Data Visualization Interpretation” (after Emory: Data Scientist at the Home Depot)
- Dr. Yanan Da (2020-2025); Thesis: “Characterizing and Mitigating Human Bias in Holistic Reviews with Interactive Visualization” (after Emory: Founding Engineer at Akify)
- Dr. Ziwei Dong (2020-2025); Thesis: “Towards Responsible Data Science with Behavior Change Interventions” (after Emory: Applied Scientist at Amazon)
- PhD Thesis Committees:
 - Dr. Aimen Gaba (2026); Thesis: “Advancing Responsible and Trustworthy AI Using Human-Centered Methods” (UMass Amherst)
 - Dr. Shrey Gupta (2024); Thesis: “Transfer and Integration of Knowledge for Irregular Spatiotemporal Data” (after Emory: Postdoctoral Scholar, Boston College)
- Master’s Thesis Committees:
 - Sunghyun Kim (2026); Thesis: “Gesturing with Paper: User-Defined Gestures and Mechanisms for Manipulating Data Visualization on Large Displays” (after Emory: PhD student, Emory)
 - Catherine Baker (2024); Thesis: “Read with Emora: Revolutionizing Children’s Reading Comprehension Through An LLM-Powered Personalized Intelligent Tutoring System for Vocabulary, Grammar, and Context-Based Learning”
- Undergraduate Honors Advisees:
 - Tommy Skodje (2024); Thesis: “An Eye-Tracking Toolkit for Evaluating Memory in Epilepsy Patients”
- Undergraduate Honors Thesis Committees:
 - Junzhi (Molly) Han (2025); Thesis: “Reducing Cognitive Load in Digital Reading: An LLM-Powered Approach for Universal Reading Comprehension”
 - Sophia Whitmire (2025); Thesis: “Friction by Design: Restoring Agency in the Attention Economy”
 - Yijun Liu (2024); Thesis: “CodeCuddle: Automating Inclusive Web Design with Large Language Model”
 - Cole Smith (2024); Thesis: “Urinary RNA Biomarkers in Extracellular Vesicles from Post-DRE Urine to Predict Aggressive Prostate Cancer”
 - Elijah Chou (2023); Thesis: “Measuring Creativity in Computer Programming: A Code Distance Approach”
 - Chen Gong (2023); Thesis: “Evaluating Speaker Diarization in Transcripts: A Text-based Approach with the TDER Metric and the TranscribeView System”
 - Alexandra Li (2023); Thesis: “Signaface: Face Generation for SSL/TLS Certificate Change Detection”
 - Noah Okada (2023); Thesis: “Development of Immersive Cognitive Tasks for Use in Human Memory Research”
 - Peyton Robertson (2023); Thesis: “Vulnerability Detection: A Machine Learning Approach to Identifying Security Vulnerabilities In Code”
 - Sierra Talbert (2023); Thesis: “Human.exe: Generating Critical Theories of Humanity in Human-Computer Interaction”
 - Kevin Wu (2023); Thesis: “CONSchema: Schema Matching with Semantics and Constraints”
 - Ruby Wu (2023); Thesis: “ClusT: Interactive Visualization Tool for Deep Constrained Clustering on Tweets”
 - Zach Zaiman (2023); Thesis: “AudioStrike: Acoustic Identification of Keystrokes to Enhance End-to-End Session Integrity”
 - Christina Chance (2022); Thesis: “Accuracy of African American Vernacular English for Closed Captioning on Zoom”

INVITED TALKS

- **Awareness, Behavior, Decisions, Oh My! Detecting and Mitigating Biases in Visualization**
 - *March 2026.* University of Maryland.
 - *January 2026.* Clemson University.
 - *October 2025.* TU Wien.
 - *May 2025.* Focus Period Linköping University: Visualization-Empowered Human-in-the-Loop Artificial Intelligence.
- **Improving Decision Making with Visualizations for Social Good**
 - *July 2025.* Gordon Research Conference: Visualization in Science and Education.
- **Information Visualization: Creating Effective Grant Figures**
 - *June 2025.* Emory CFDE NSF CAREER Academy.

- **A Visualization Viewpoint on Mitigating Metacognitive Deficits**
 - *October 2024.* Human-Centered Computing (HCC) Seminar at University of Minnesota
 - *March 2024.* Scientific Computing and Imaging (SCI) Institute at University of Utah.
 - *September 2022.* Emory SON Center for Data Science.
 - *September 2022.* ATLAS Institute at University of Colorado - Boulder.
- **Good and Bad Biases in Innovation, Decision Making, and Beyond**
 - *November 2023.* Emory Center for Mind, Brain, and Culture.
- **Bias in Visualization: Considering Equity through Research**
 - *November 2021.* Object Perception, visual Attention, and visual Memory (OPAM) conference
- **As We Are: Detecting and Mitigating Human Bias in Visual Analytics**
 - *April 2021.* University of Utah, Data Science Seminar. Virtual Seminar.
 - *October 2020.* Pacific Northwest National Laboratory. Virtual Brown Bag.
 - *September 2020.* Emory University. Atlanta, GA. Abstract
 - *May 2020.* CMU & AFRL, Center for Excellence on Human-Machine Teaming. Virtual Brown Bag.
 - *May 2020.* Siemens. Virtual Brown Bag.
 - *March 2020.* Cornell University, Info Science Colloquium. Ithaca, NY. Abstract
 - *February 2020.* University of Washington, Allen School Colloquium. Seattle, WA. Watch
 - *February 2020.* Emory University. Atlanta, GA. Abstract
 - *February 2020.* New York University. New York, NY. Abstract
 - *February 2020.* University of Wisconsin. Madison, WI.
- **Left, Right, and Gender: Mitigating Implicit Human Bias in Visual Analytics**
 - *January 2020.* Georgia Institute of Technology, GVI Brown Bag. Atlanta, GA
- **The Interactive Analytics Library: Treating Interaction as Data**
 - *May 2017.* Pacific Northwest National Laboratory. Richland, WA.

PROFESSIONAL SERVICE

- Organizing Committee
 - VIS 2026: Short Papers Chair, Ombuds Committee
 - VIS 2024-2025: Doctoral Colloquium Chair
 - VIS 2021-2023: Elections Chair
- Associate Editor
 - 2024-present: Journal of Visualization and Interaction (JoVI)
 - 2025-present: Transactions on Visualization and Computer Graphics (TVCG)
- Program Committee Member
 - IEEE VIS: 2021-2023, 2025
 - EuroVis: 2022, 2024, 2025
 - EuroVis Short Papers: 2026
 - ACM CHI: 2022, 2024
 - ACM FAccT: 2023
 - BELIV Workshop on "Designing and Evaluating Visualizations for an Ethical, Inclusive, and Responsible Future": 2022
 - Vis4Good Workshop at IEEE VIS: 2021
 - IEEE VIS Short Papers: 2020
 - VISxAI Workshop at IEEE VIS: 2019
 - CHI Late Breaking Work: 2018
- College Board AP Computer Science Fellow: 2025
- Organizer
 - Visualization for Social Good Meetup at IEEE VIS 2024Visualization for Social Good Workshop at IEEE VIS 2023
 - Visualization for Social Good Workshop at IEEE VIS 2022
 - TRust and EXpertise in Visual Analytics (TREX) Workshop at IEEE VIS 2021
 - TRust and EXpertise in Visual Analytics (TREX) Workshop at IEEE VIS 2020
- Panelist
 - Establishing and Thriving in an Academic Career @ IEEE VIS, 2023
 - What is a Visual Analytics contribution, and how is it changing? @ IEEE VIS, 2023
 - Women in Tech Panel, Emory Data Science Club and ProgramHers, 2023
 - Equity in Vision Science at Object Perception, visual Attention, and visual Memory (OPAM) conference, 2021
 - Advancing Visualization Inclusion and Diversity (AVID) Workshop at IEEE VIS, 2017

INSTITUTIONAL SERVICE

- Graduate Admissions Committee (Emory University, 2020-2026)
- Faculty Search Committee (Emory University, 2020-2021, 2025-2026)
- Honors Program Selection Committee (Emory University, 2023-2026)
- Panelist:
 - DEI Statement Panel, Emory University, 2021
 - Innovations in Online Teaching Panel, Emory University, 2021